

3.6 Directional Derivatives and Gradients Worksheet

1. True or False: The directional derivative of $f(x, y)$ at the point (x_1, y_1) in the direction of θ_1 is given by $f_x(x_1, y_1)\sin(\theta_1) + f_y(x_1, y_1)\cos(\theta_1)$.

2. Find the directional derivative of $f(x, y, z) = 3x^2y^2 + 2yz$ at the point $(-1, 0, 4)$ pointing in the direction $\langle -1, 3, 3 \rangle$.

3. Let $f(x, y) = 20 - (2x^2 + y^2)$.
 - a) Find a vector at the point $(1, 3)$ pointing in the direction in which f is increasing fastest?

 - b) What is the maximum rate of change?

4. Let $f(x, y) = xye^{x^3+x+y^2}$.

a) Find ∇f .

b) Find a vector that is orthogonal to the curve $f(x, y) = e^3$ at the point $(1, 1)$.

5. Let $\mathbf{u} = \langle 1, 2, -1 \rangle$. Let $f(x, y, z) = \tan^{-1}(xy) + e^{z^2+x}$. Find the derivative of f in the direction of $\mathbf{u}$ at the point $(1, 1, 1)$.

6. Find a unit vector which is normal to the surface $z = x^2y^2 + y + 1$ at the point $(0,0,1)$.

7. Let $f(x,y,z) = \frac{x}{y} - yz$. Find the directions in which $f(x,y,z)$ increases and decreases most rapidly at the point $(4,1,1)$, and then find the directional derivatives of $f(x,y,z)$ along these directions.

8. Let $f(x, y, z) = 2ye^{x-2y} + \ln(z)$

a) Find the directional derivative of $f(x, y, z)$ at the point $(2, 1, 1)$ along the direction of the vector $\langle -3, 1, 2 \rangle$.

b) Is the direction derivative in part(a) maximum among all directional derivatives of $f(x, y, z)$ at $(2, 1, 1)$?
If not, find the maximum one and the direction along which it appears.

c) Find a direction $\mathbf{v}$ such that $D_{\mathbf{v}}f(2, 1, 1) = 0$.

9. The temperature in degrees Celsius on the surface of a metal plate is $T(x, y) = 20 - 4x^2 - y^2$ where x and y are measured in centimeters. In what direction from the point $(2, -3)$ does the temperature decrease most rapidly? What is this rate of decrease?